Ward 3D Handover App – System & Information Flow

Nov 7, 2026 · @Richard

Overview

The Ward 3D handover app replaces the paper booklet with a signed, auditable digital handover for the nurse in charge (NIC), and gives the ward manager a view across shifts. This document follows one piece of information from the moment a nurse requests access to the moment it appears on an admin screen.

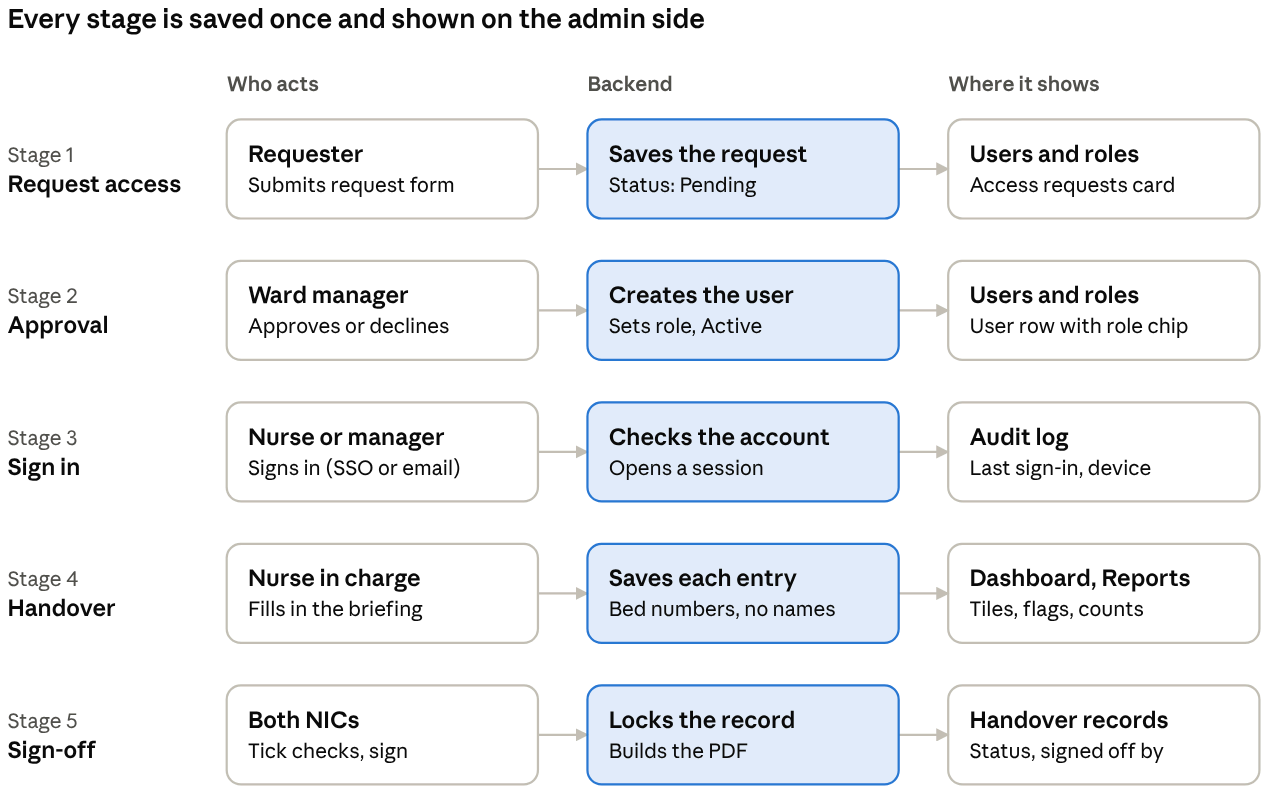
The system has three layers:

- **Screens.** What people see: the sign-in and request-access pages, the four handover screens used by nurses, and the seven admin screens.
- **Backend.** The service that checks who you are, validates and saves what you type, builds the PDF and writes the audit log.
- **Data.** The database tables that hold users, handover records, concerns, lines and feeds, medications, and audit events.

Five kinds of people use it: ward staff (read the current handover), the NIC (fills in and signs off), the ward manager (approves users, reviews records and reports), the admin (manages users and form settings) and the information governance (IG) auditor (reads the audit log only).

The screens in this document are design mock-ups with sample data. The hosting platform is not yet chosen, so the backend is described by what it does, not by product.

End-to-end flow at a glance



end-to-end flow · 5 stages, 3 layers

Read each row left to right: a person acts, the backend (shaded) saves it once, and the same record then appears on an admin screen. The stages run in order, from request to sign-off.

Stage 1 – Sign-up (Request access)

Sign-up creates a pending request, not a working account: nobody can reach patient information until the ward manager approves them. The page is shown to anyone who opens the app and chooses "Request access" from the sign-in screen.

Field entered	Rule checked by the backend	Saved to
Full name, staff number	Required; staff number must be unique	Access request
NHS email	Must be a Trust (nhs.net) address; one request per email	Access request

Field entered	Rule checked by the backend	Saved to
Role requested	One of: Ward staff, NIC, Ward manager	Access request
Ward	Ward 3D	Access request
Line manager / approver	Must be an existing ward manager or admin	Access request
Password and confirmation	12+ characters, upper and lower case, a number, a symbol; stored only as a hash	Credentials (not shown anywhere)
IG training and confidentiality tick-boxes	Both must be ticked	Access request (with date and time)

What the backend does. It validates the form, saves the request with the status Pending, and notifies the chosen approver. It writes an audit event, "Access requested", with the email and the device.

What is displayed, and where.

- To the requester: a confirmation page saying the request is awaiting approval. The password is never shown again.
- To the approver: a new row in Admin > Users & roles > Access requests, and the Pending count on that card ("2 waiting").
- To the IG auditor: the audit event in Admin > Audit log.

Stage 2 – Approval on the admin side

A request becomes an account only when a ward manager or admin approves it in Users & roles > Access requests. This is the first point where information from the public-facing screens appears on the admin side.

What the admin sees	Source	Action available
Name and unit (for example "N. Osei · new starter")	Access request: name, role, line manager	Approve, Decline
Role asked (NIC, Ward staff)	Access request: role requested	Approve (the admin can change the role first)

What the admin sees	Source	Action available
"2 waiting" badge	Count of requests with status Pending	None

On Approve. The backend creates the user, links it to the Trust login, sets the role and the status Active, and tells the person they can sign in. The row moves from Access requests to the Users list, with Last sign-in empty.

On Decline. The request is closed with the reason and the approver's name; the person is told, and no account is created.

What is logged. Both outcomes write an audit event (who decided, which request, the role granted), shown in Admin > Audit log as "Role changed" or a decline entry.

Stage 3 – Sign in

Sign-in proves who the person is and decides which screens they get. The first choice is the Trust single sign-on button; email and password is the fallback.

1. The person chooses single sign-on, or enters their NHS email and password.
2. The backend checks the account exists, is Active and has not been disabled. Three failed attempts are recorded as "Failed sign-in".
3. Admin and manager roles are asked for a second factor (two-factor sign-in is switched on in Data & retention).
4. The backend opens a session and reads the user's role. The role decides the landing page: the handover form for nurses, the Dashboard for managers and admins.
5. The session locks after the inactivity period set in Data & retention (2 minutes in the sample), and the user signs in again.

What is displayed, and where.

- To the user: their name and role in the app header, and for managers the "Back to handover form" button in the admin menu.
- On the admin side: the user's Last sign-in time in Users & roles, and a "Sign-in" event in Audit log with the device (for example "Ward tablet 1").
- On the Dashboard: three failed sign-ins appear under Needs attention as a security item.

"Keep me signed in" is off by default because ward tablets are shared.

Stage 4 – The handover (nurse screens)

The NIC chooses "Start new handover" on the cover; the backend creates one handover record for that date and shift with the status In progress. Every change on the next three screens is saved to that record as it is entered, so a lost connection does not lose the briefing. Patients are identified by bed number, not by name.

Section (screen)	What the NIC enters	Saved as	Displayed afterwards
Header (all screens)	Date, shift, nurse in charge	Handover record header	Admin: Handover records list; Dashboard "Today's handovers by shift"
Patient & staffing overview (2)	Total, empty beds, amber, for discharge, staffing concerns	Handover record: counts	Dashboard (Pts, Amber); Reports (patients per shift, occupancy)
Unwell / deteriorating, NEWS (2)	Bed, NEWS score, action	Patient concern rows of type NEWS	Dashboard: "Deteriorating (NEWS)" tile and Open safety flags; Handover records: Flags column
Equipment & safety checks (2)	Resus trolley used, checked and sealed, other equipment issues	Handover record: checks	Sign-off checklist (Screen 4); the printed PDF
Appointments, transport, mobility, clinical and surgical instructions (2)	Free text and level or duration	Handover record: text fields	The printed PDF; the record opened from Handover records
Clinical & safeguarding concerns, 17 types (2)	Tick a concern, then bed and notes or action	Patient concern rows: bed, type, note, action	Dashboard: DNAR and Safeguarding tiles, Open safety flags; Reports: Most frequent concerns (counts only)

Section (screen)	What the NIC enters	Saved as	Displayed afterwards
Pre & post surgery, labs, infusion, drain (3)	NBM, imaging, theatre, TCI, blood and IV entries, drain notes	Surgical and labs rows	The PDF and the record view for the next NIC
Vascular access and external feed (3)	CVC and PICC lines, NG, NJ, RIG, TPN, PEG, J-tube, notes	Lines and feeds rows	The PDF and the record view; External Feed can be hidden in Form settings
Governance & safety (3)	Complaint, incident report, Datix reference, IDH and POC	Governance rows	Reports: Incidents (Datix), Complaints, Datix incidents by category
Medications, TTA, transport (3)	Controlled drug (CD) and Schedule 3 status, TTAs outstanding, hospital transport reference	Medication and TTA rows	Dashboard: "TTAs outstanding" tile and Needs attention; the PDF
Additional note (3)	Free text	Handover record: note	The PDF and the record view

The ward values on Screen 4 are display only. Nothing on that page is saved apart from the sign-off, which is covered in Stage 5.

Stage 5 – Sign-off and locking

A handover is complete only when the outgoing and incoming NIC have both signed, and from that moment the record cannot be edited. This is the step that turns a form into a record the ward manager can rely on.

1. On Screen 4 the NIC ticks the six checks: NEWS patients discussed, DNAR and palliative confirmed, safeguarding handed over, resus trolley checked and sealed, controlled drugs checked, TTAs and transport noted.

2. The outgoing NIC enters their name and the time; the incoming NIC signs in on the same screen with their own login. "Require incoming NIC sign-off" is switched on in Data & retention.
3. The backend refuses sign-off while a check is unticked or a signer is missing, then sets the status to Signed off and locks the record.
4. The backend builds the PDF and keeps it with the record. The NIC can print or save it with "Print / Save PDF".
5. The incoming shift sees the signed record as the current handover, read-only. Later changes need a new record.

What is displayed, and where.

- Dashboard: the shift changes from In progress to Signed off, with "Signed off by" and the time.
- Handover records: the status chip and the Signed off by column, and an Open and PDF button on the row.
- Audit log: "Signed off" and "Printed" events.
- If nobody signs, the record stays "Not signed off" in red, and the Dashboard counts it under Unsigned handovers and Needs attention.

Backend: services and data

The backend sits between the screens and the database, and every screen reads and writes through it, never to the database directly. It does four jobs: identity, validation, record-keeping and reporting.

Services

- **Identity.** Links to Trust single sign-on, holds roles, checks the second factor, and starts and locks sessions.
- **Handover service.** Validates each field, saves it to the open record, enforces the sign-off rules and locks signed records.
- **Form configuration.** Serves the current version of the form (concerns, sections, dropdowns) to the nurse screens, and keeps older versions so past records still display as they were filled in.
- **Reporting.** Counts concerns, incidents and completion by shift and period, using bed numbers and counts only, and builds the PDF and Excel exports.

- **Audit and retention.** Writes one audit event for every view, edit, sign-off, print, export and settings change, and deletes records when their retention period ends.

Tables

Table	Holds	Written by	Read by
Users	Name, staff number, email, role, status, last sign-in	Approval of an access request; admin edits	Sign-in; Users & roles
Access requests	Request fields, status, approver, decision	Request access page; approver	Access requests card
Handover records	Date, shift, NIC, counts, checks, notes, status, signers	Nurse screens 2 to 4	Dashboard, Handover records, Reports, PDF
Patient concerns	Record, bed, concern type, note, action	Screen 2	Dashboard flags, Reports
Surgical, labs and governance rows	NBM, blood, IV, Datix reference, complaint, drain	Screen 3	Reports, PDF
Lines, feeds, medications and TTA	CVC, PICC, NG, NJ, CD status, TTA outstanding	Screen 3	Dashboard TTA tile, PDF
Form configuration	Concerns, sections, dropdown lists, version number	Form settings	Nurse screens
Audit events	Time, user, action, record, device	Every service	Audit log
Settings	Retention, auto-lock, export rules, two-factor	Data & retention	All services

Passwords, if email sign-in is used, are stored only as salted hashes. With single sign-on the Trust holds them.

Admin side: where information is displayed

Each of the seven admin screens is a different view of the same tables the nurses write to. Nothing on the admin side is typed twice: it is read from the records, concerns, users and audit events described above.

Admin screen	What it displays	Comes from	Who can open it
Dashboard	Handovers per shift, tiles for DNAR, NEWS, safeguarding, TTAs and unsigned handovers, open safety flags by bed, 7-day completion, Needs attention, recent activity	Handover records, patient concerns, medication and TTA rows, audit events	Ward manager
Handover records	Searchable list of past handovers with status, signers and flag counts; Open and PDF per row	Handover records, locked PDFs	Ward manager
Users & roles	Staff list with role and last sign-in, the permissions grid, access requests	Users, access requests	Admin
Form settings	Concerns and sections with on/off switches, dropdown lists, version history	Form configuration	Admin
Reports	Completion rate, incidents, complaints, average patients per shift, most frequent concerns, Datix by category, weekly completion; PDF and Excel export	Aggregated counts from handover records and governance rows	Ward manager
Audit log	Time, user, action, record and device for every event; export	Audit events	Ward manager, admin, IG auditor

Admin screen	What it displays	Comes from	Who can open it
Data & retention	Retention period, auto-lock, export and name rules, scheduled deletions, compliance checklist	Settings, record dates	Admin

Three behaviours connect the nurse side to these screens:

- **Live updates.** A concern ticked on Screen 2 changes the Dashboard tiles and flags on the next refresh, because both read the same rows.
- **Changes flow the other way.** Switching a concern off in Form settings removes it from the nurse form from the next handover; earlier records keep it.
- **No editing of history.** Handover records and the Audit log are read-only on the admin side.

Who sees what

Only the NIC can write to a handover, and only the NIC, ward staff and ward manager can read patient notes. The admin and the IG auditor see how the system is used, not what is written about patients.

Permission	NIC	Ward staff	Ward manager	Admin	IG auditor
Fill in and sign off a handover	Yes	No	No	No	No
View the current handover	Yes	Yes	Yes	No	No
View past records	No	No	Yes	No	No
See patient notes	Yes	Yes	Yes	No	No
Reports	No	No	Yes	No	No
Manage users	No	No	No	Yes	No
Edit form settings	No	No	No	Yes	No
View the audit log	No	No	Yes	Yes	Yes

Rules that apply to every screen:

- **Bed numbers, not names.** Handover screens, the Dashboard and the PDF identify patients by bed number when "Use bed numbers instead of names" is on.

- **Counts only in reports.** Report exports hide patient identifiers by default, so Reports and its Excel and PDF files show totals, never rows about one patient.
- **The audit log holds events, not notes.** It records that a record was opened or printed, not what it said.
- **Role comes from the account.** A role is set only through approval or an admin edit, and each change is an audit event.

The Dashboard and Handover records show bed-level flags, so they sit with the ward manager. Whether the admin also gets a Dashboard without flags is an open choice, listed at the end.

Audit, retention and governance

Every action that touches a record or a setting leaves an audit event, and records are deleted automatically when their retention period ends. Both are visible to the right people on the admin side, so the ward can show who did what and when.

Event	Written when	Shown in
Sign-in, Failed sign-in	A user signs in, or three attempts fail	Audit log; Dashboard security item; Last sign-in in Users & roles
Opened, Edited	A record is viewed or changed	Audit log
Signed off, Printed, Exported	A handover is signed, printed, or a report or PDF is exported	Audit log; Handover records status
Role changed	An access request is approved or a role is edited	Audit log; Users & roles
Form changed	Form settings are published as a new version	Audit log; Form settings version history
Viewed log	Someone opens the audit log	Audit log

Retention. The retention period is set in Data & retention (the mock-up shows 8 years as a sample value; the real figure follows Trust policy). The backend deletes records past that date, and the screen lists the next scheduled deletions with their counts, so nothing is removed without being visible first.

Governance before real patient data is used. The Compliance card in Data & retention tracks the DPIA, information governance sign-off and the clinical safety case (DCB0129 and DCB0160), plus any data sharing agreement and penetration test the Trust asks for. The items marked Done in the mock-up are placeholders.

Open decisions

Six choices change how the flows above are built, and each needs an answer from the ward or the Trust digital team before development starts.

- [ ] **Platform.** Power Apps with SharePoint or Dataverse, or a custom web app. First check whether the Trust's electronic patient record already has a handover module.
- [ ] **Patient identification.** Bed number only, or bed number plus initials, on handover screens and the PDF.
- [ ] **Incoming NIC sign-off.** Signing on the same tablet with their own login, or from their own device.
- [ ] **Admin view.** Whether the admin gets a Dashboard without bed-level flags, or only the manager sees it.
- [ ] **Approver.** Whether the ward manager, the admin or both can approve access requests.
- [ ] **Retention period.** The real figure under Trust policy, replacing the 8-year sample.